Prescribed chemistry in Ciais et al. (2026)

Reference. Ciais et al., *Science* 391, eadx8262, doi:10.1126/science.adx8262. The full article and 46-page supplement (Figs. S1–S18, Table S1) were read. Numerical comparisons retain 2019–2022.

Method. Three GOSAT inversions prescribe two independently constructed OH fields. OH-1 combines observed precursor concentrations with grid-cell chemical-box sensitivities evaluated for 2005. OH-2 uses ERA5-driven LMDz–INCA chemistry with updated precursor emissions, including aviation; its magnitude is multiplied by 0.88 to match the earlier TRANSCOM reference. OH-1 assumes fixed NO_x vertical profiles. PYVAR–LMDz optimizes weekly total fluxes at $1.9^\circ \times 3.75^\circ$; WESTLAKE optimizes seasonal land fluxes at $4^\circ \times 5^\circ$; GONGGA optimizes four source categories at $2^\circ \times 2.5^\circ$ in 14-day windows, with an initial-state adjustment using the first seven days. Wetland priors have no interannual variation. Quoted spreads combine three models and two OH fields, without sampling all uncertainty in the prescribed chemistry.

Findings. Emissions increase by 22.0 ± 3.5 Tg CH_4 yr^{-1} in 2020 relative to 2019. Wetland and inland water emissions average 8.6 ± 2.6 Tg CH_4 yr^{-1} above 2019 in 2020–2022. Africa and Southeast Asia dominate the increase; southern South American wetlands instead decrease by 1.4 ± 0.8 Tg CH_4 yr^{-1}. The headline 83% OH contribution uses 2019–2023, outside our target interval. The supplement defines it as the mean of four fractions of adjacent-year growth changes explained by changes in OH removal. It is neither a fraction of cumulative growth nor a fixed-2019 forcing fraction. Separate reported OH–growth regression R^2 values are 0.5–0.7.

Sector interpretation. In mixed grid cells, emission changes are reallocated when the agricultural share differs by more than 20% from the corresponding bottom-up fraction. Thus wetland versus agricultural attribution is partly prescribed. Prior-proportional allocation would assign $46.8 \pm 14.8\%$ of global changes to agriculture/waste and $34.7 \pm 4.6\%$ to wetlands. A global isotope-box calculation favors GONGGA's source mixture, conditional on source signatures and fractionation assumptions. The supplement inconsistently describes GONGGA's treatment of other natural sources on pp. 4–5; those components are not independently reconstructed here.

Data and comparison. Table S1 supplies bottom-up baseline levels and period-mean anomalies. Annual inversion emissions and OH-removal anomalies are available graphically. We digitize central values and separate reading precision from ensemble spread. Native annual burdens and absolute 2019 OH removal are not tabulated. Conversion to He's OH forcing therefore uses an explicit common-burden/lifetime scenario. The air-mass-weighted OH maps cannot replace reaction-weighted loss frequencies. No inversion is repeated.